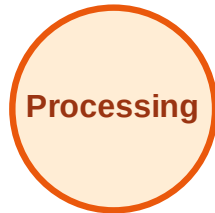


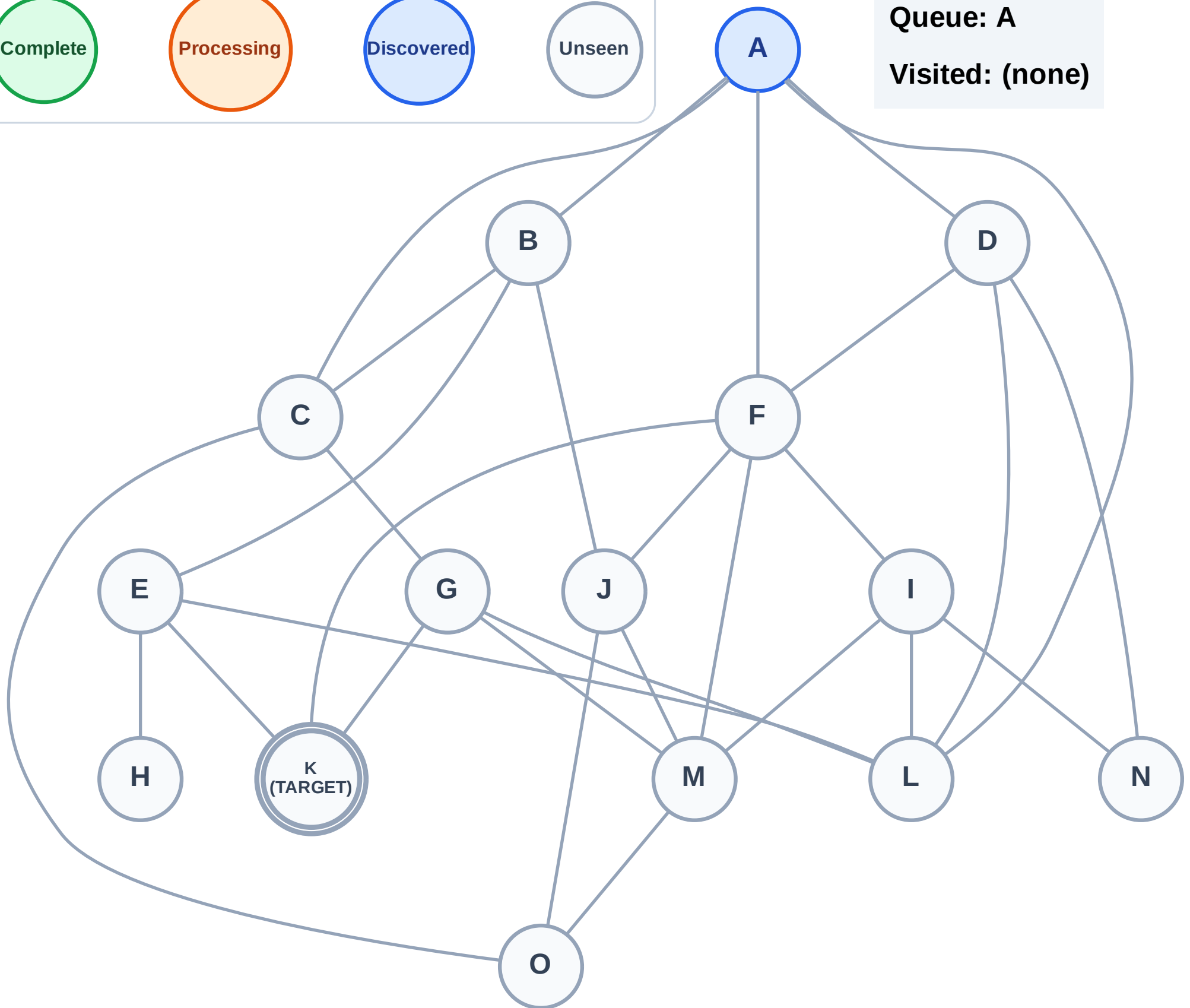
BFS Traversal

Step 1: Start at A (Seeking target node K (Eager Short-Circuit))

Legend



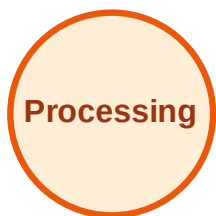
Queue: A
Visited: (none)



BFS Traversal

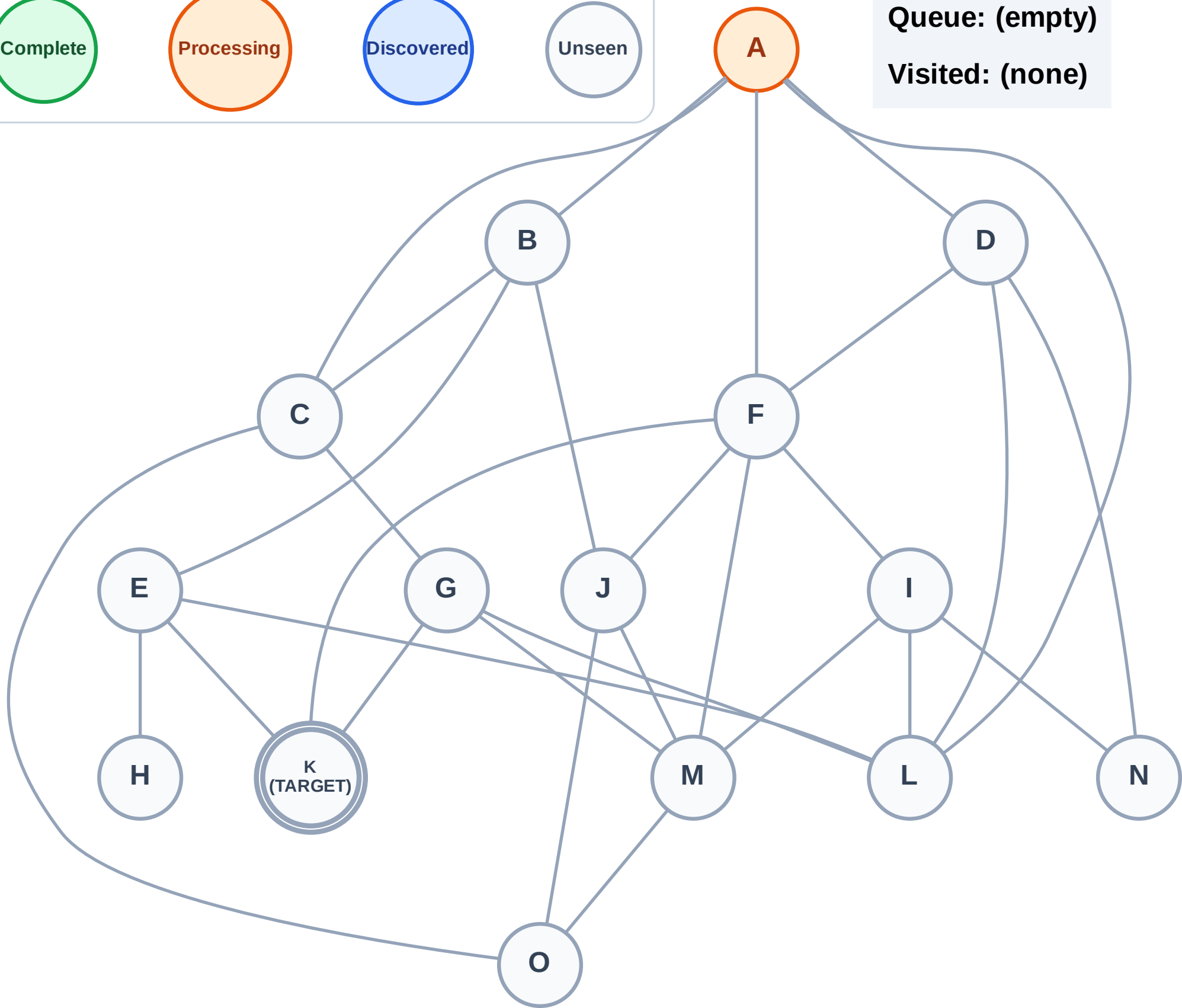
Step 2: Process node A

Legend



Queue: (empty)

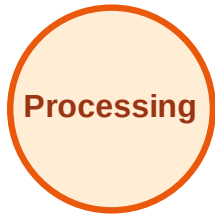
Visited: (none)



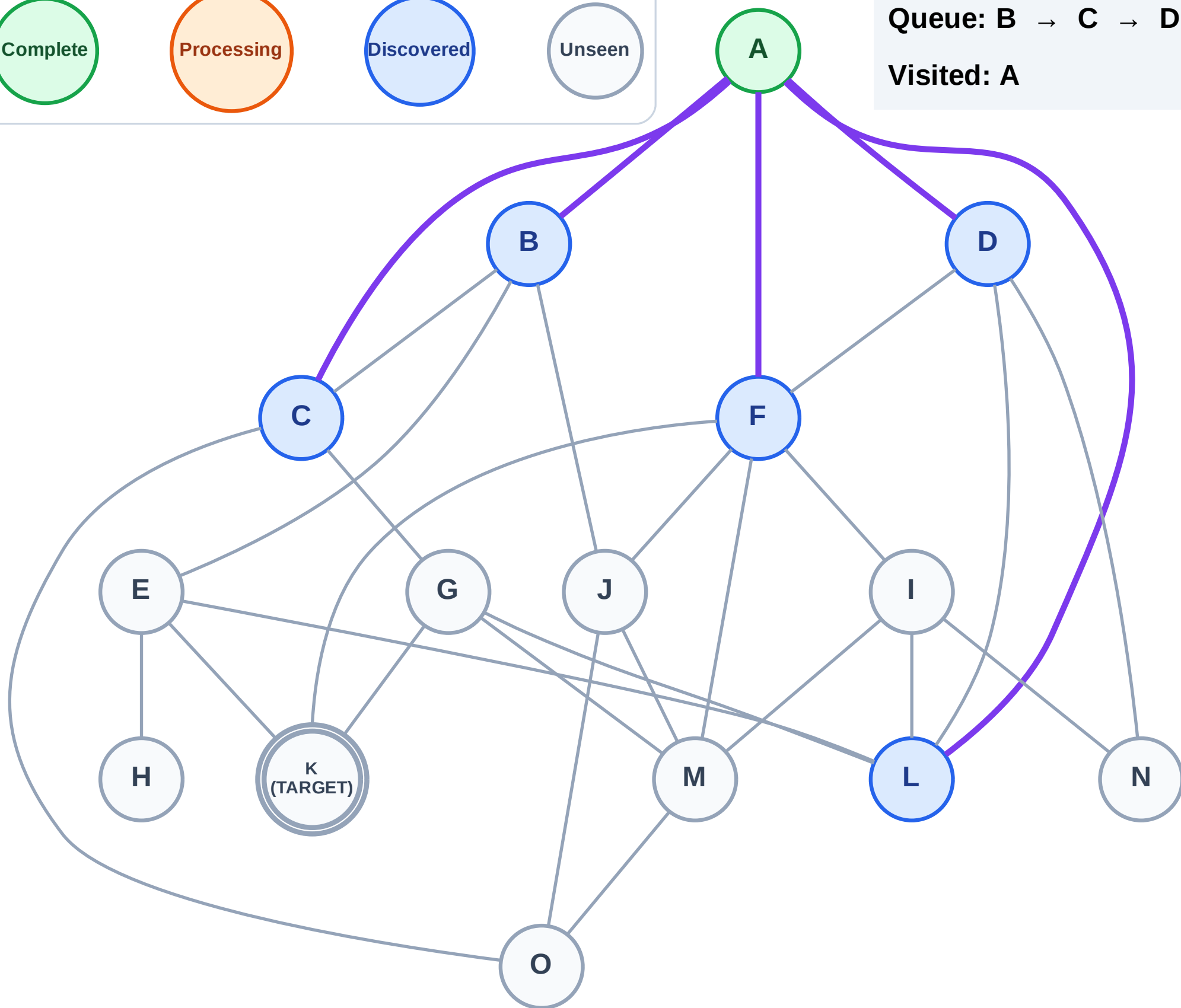
BFS Traversal

Step 3: Discover B, C, D, F, L from A

Legend



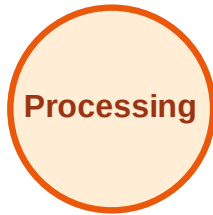
Queue: B → C → D → F → L
Visited: A



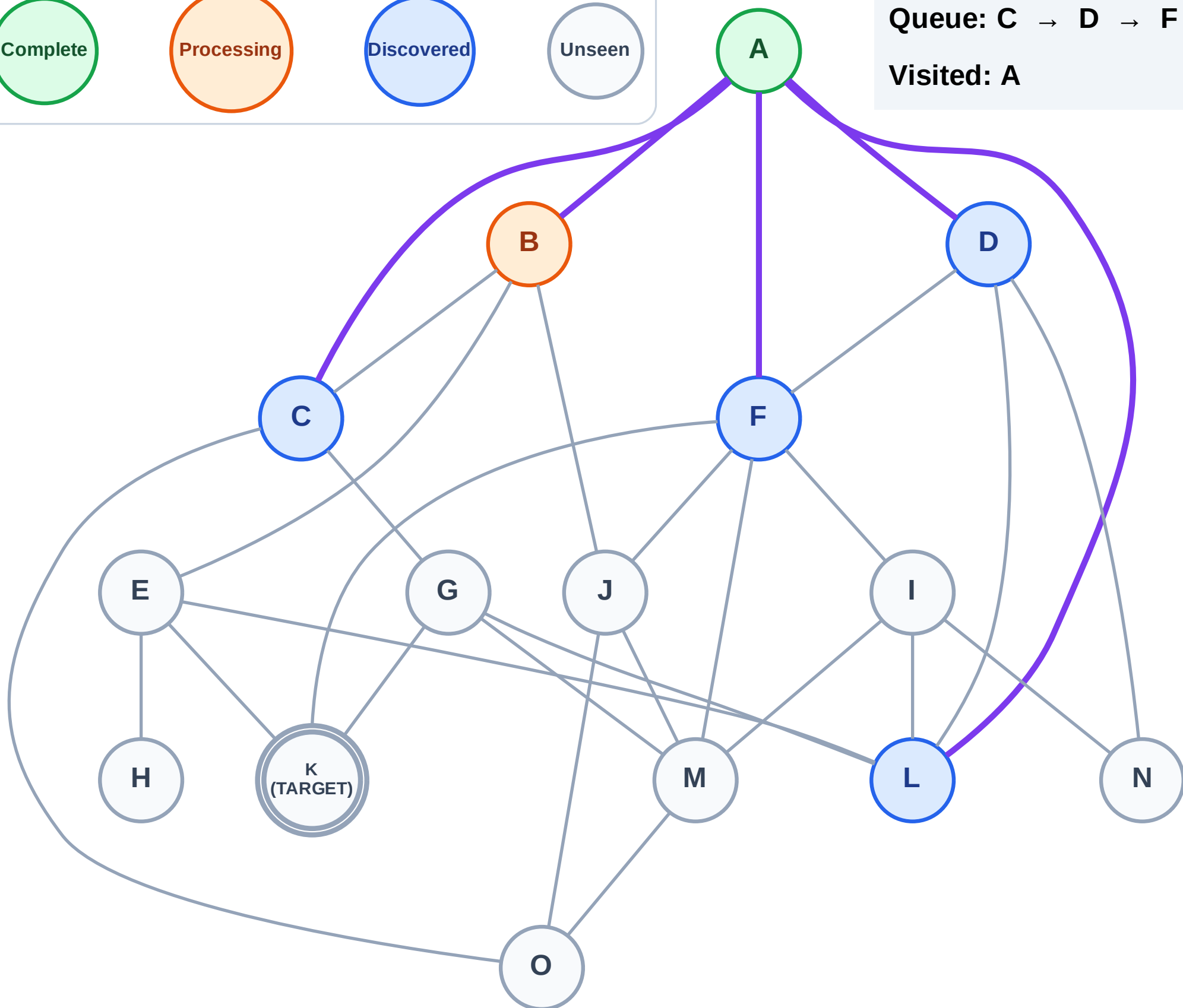
BFS Traversal

Step 4: Process node B

Legend



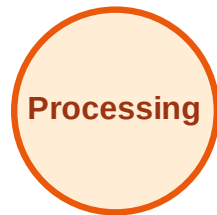
Queue: C → D → F → L
Visited: A



BFS Traversal

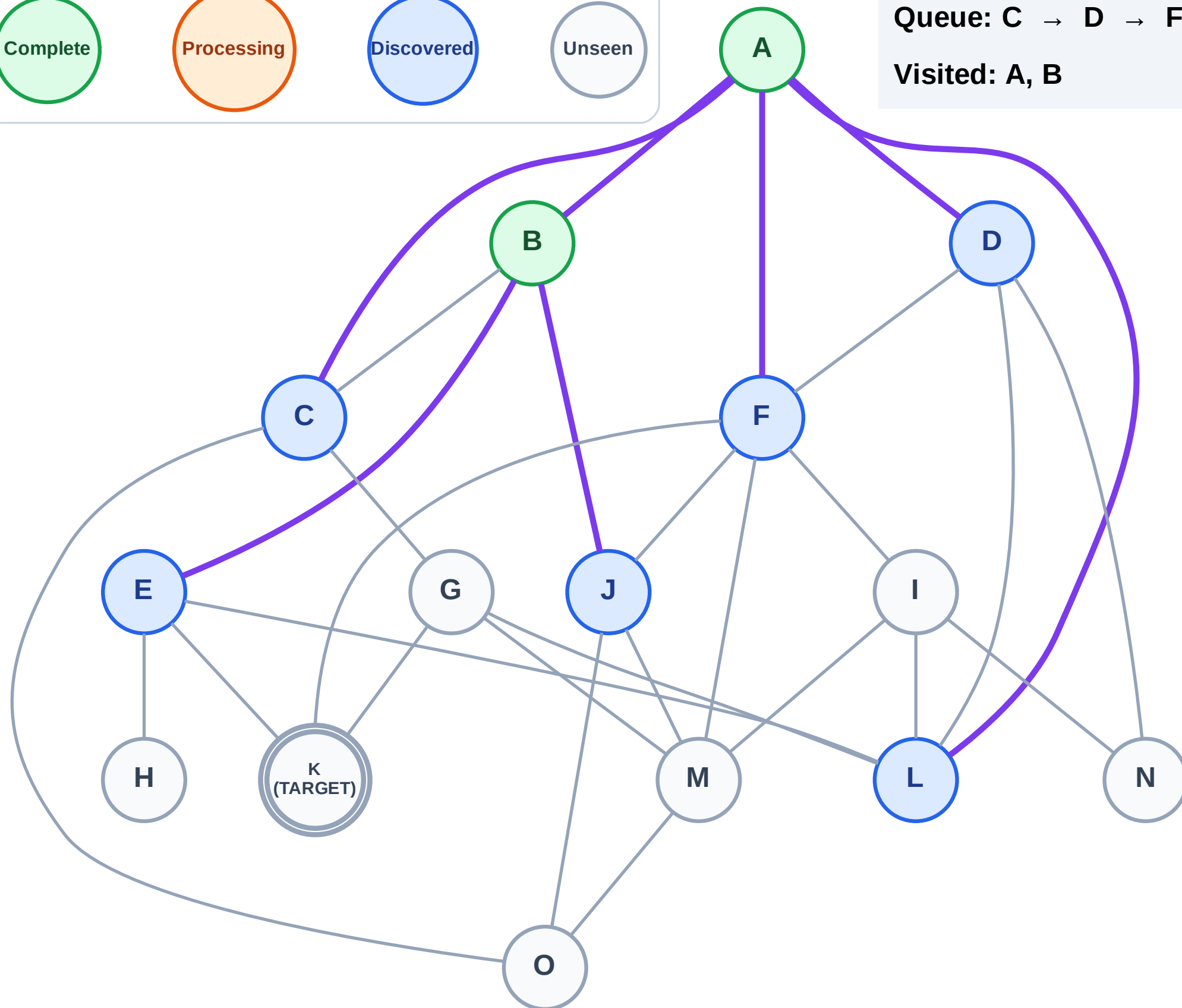
Step 5: Discover E, J from B

Legend



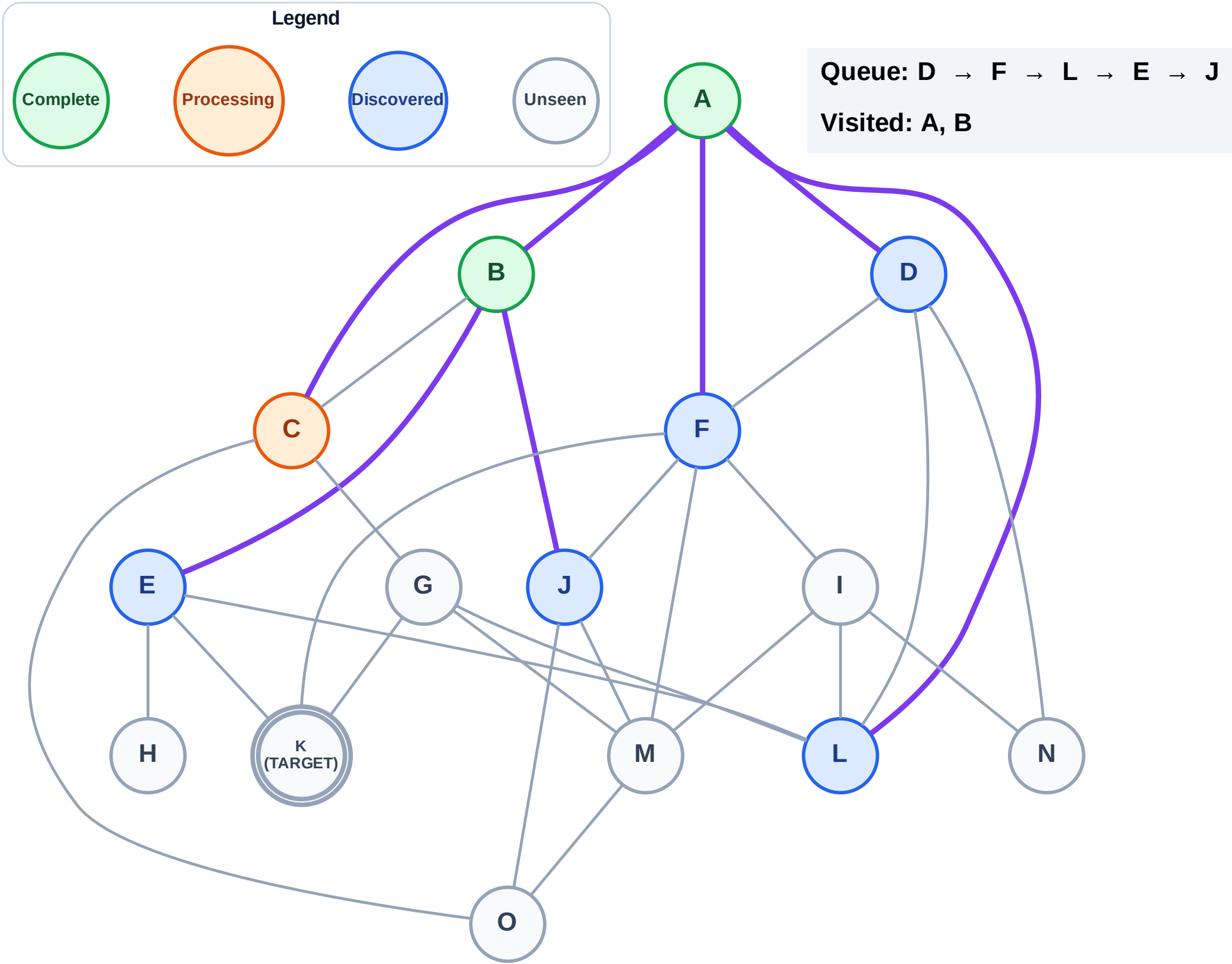
Queue: C → D → F → L → E → J

Visited: A, B



BFS Traversal

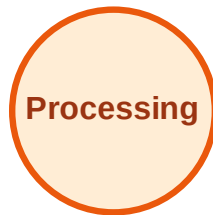
Step 6: Process node C



BFS Traversal

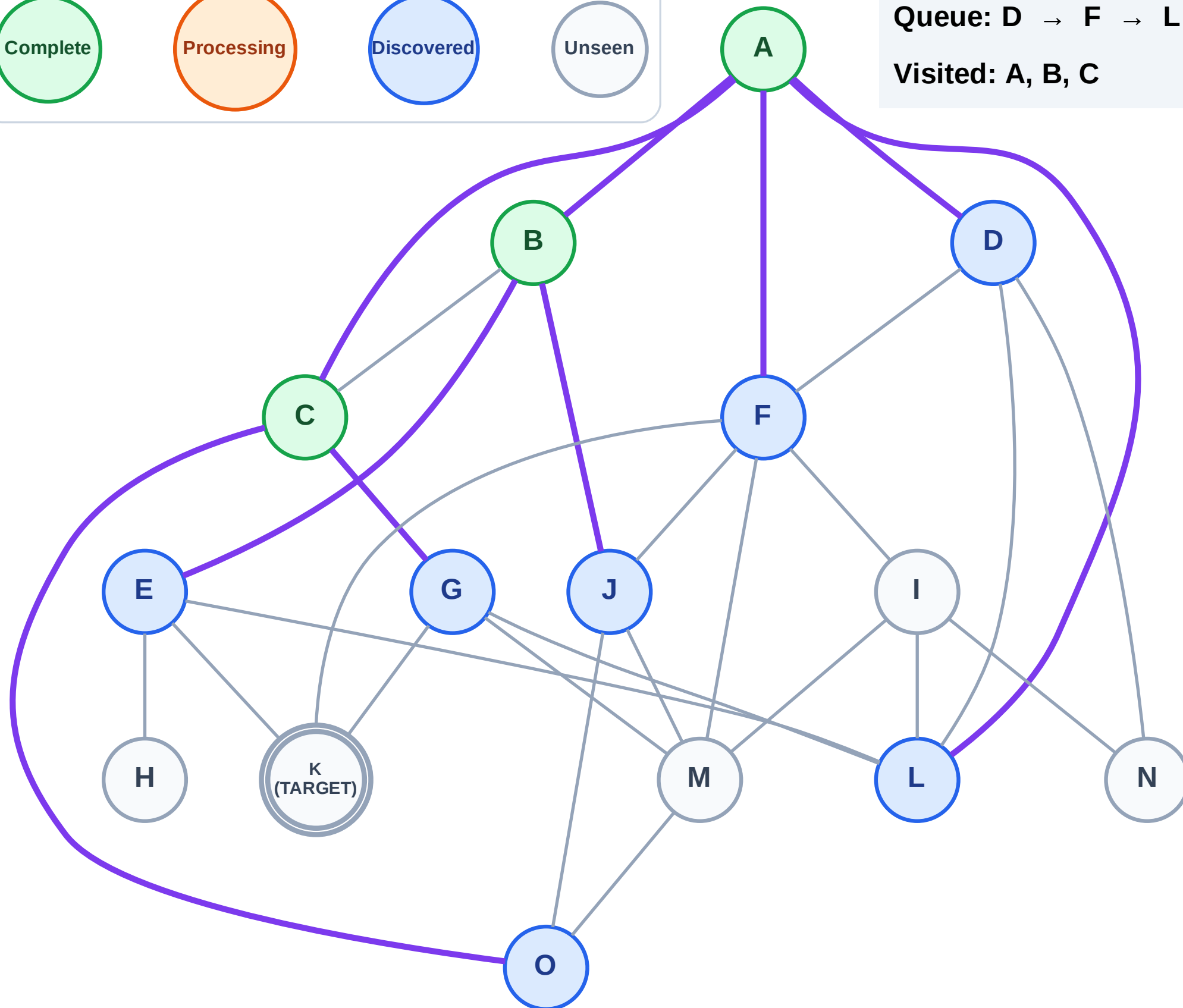
Step 7: Discover G, O from C

Legend



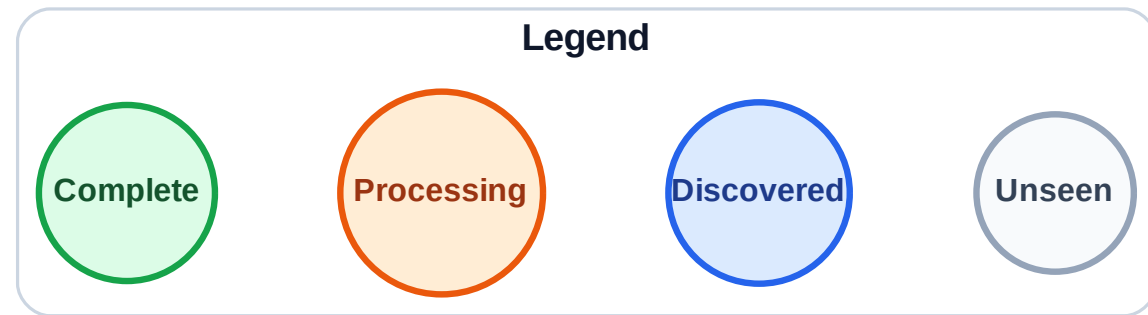
Queue: D → F → L → E → J → G → O

Visited: A, B, C



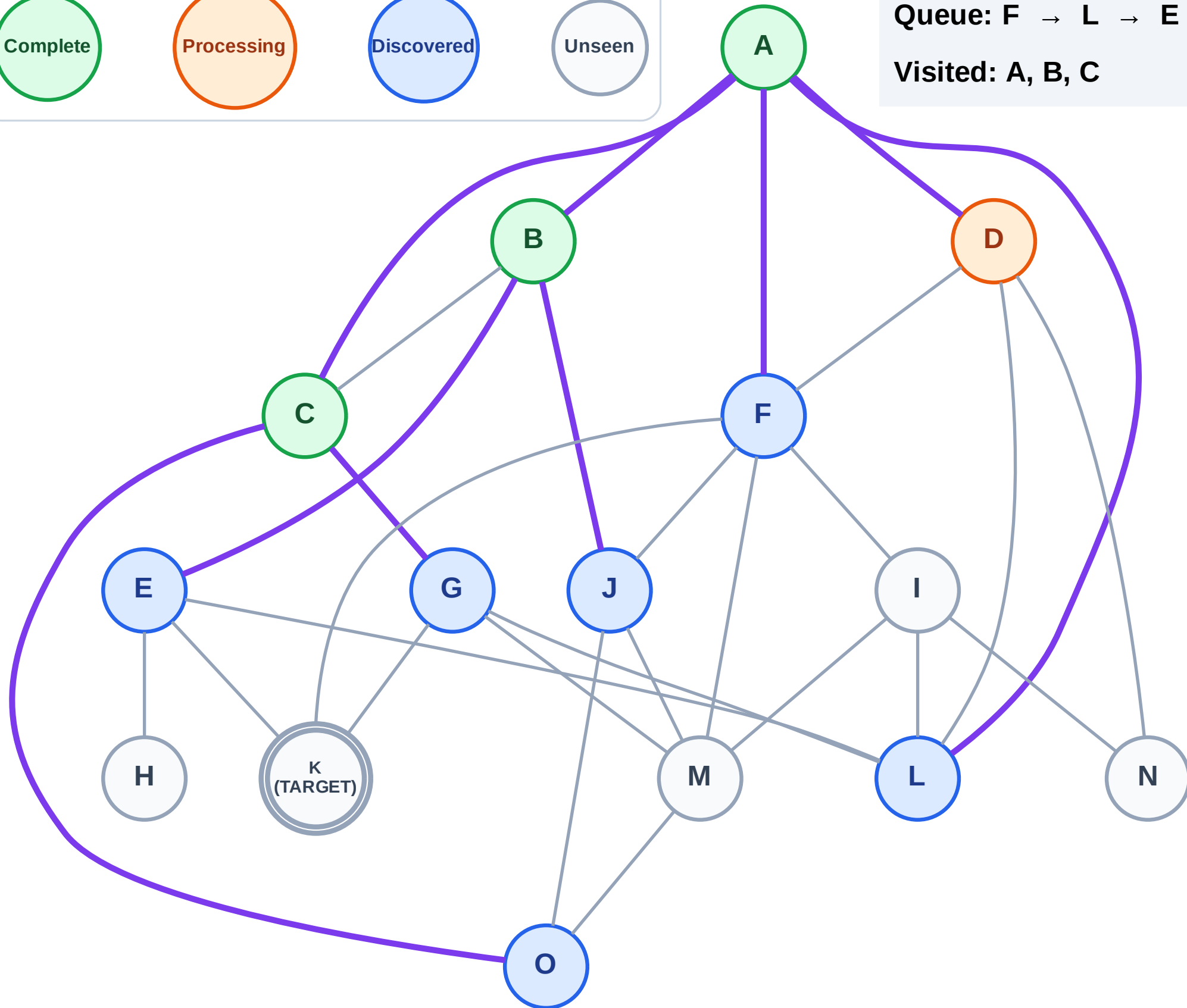
BFS Traversal

Step 8: Process node D



Queue: F → L → E → J → G → O

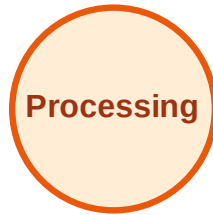
Visited: A, B, C



BFS Traversal

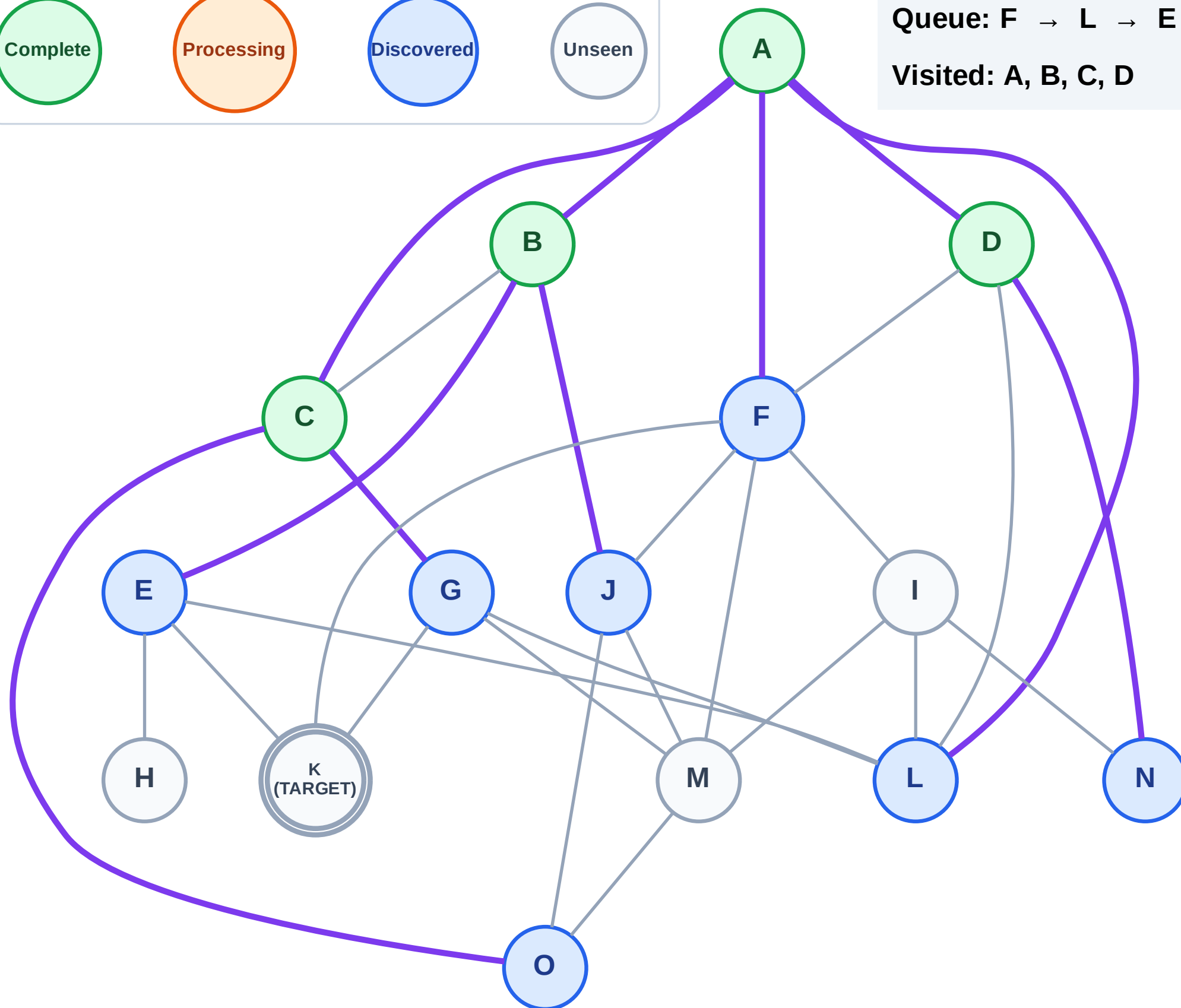
Step 9: Discover N from D

Legend



Queue: F → L → E → J → G → O → N

Visited: A, B, C, D

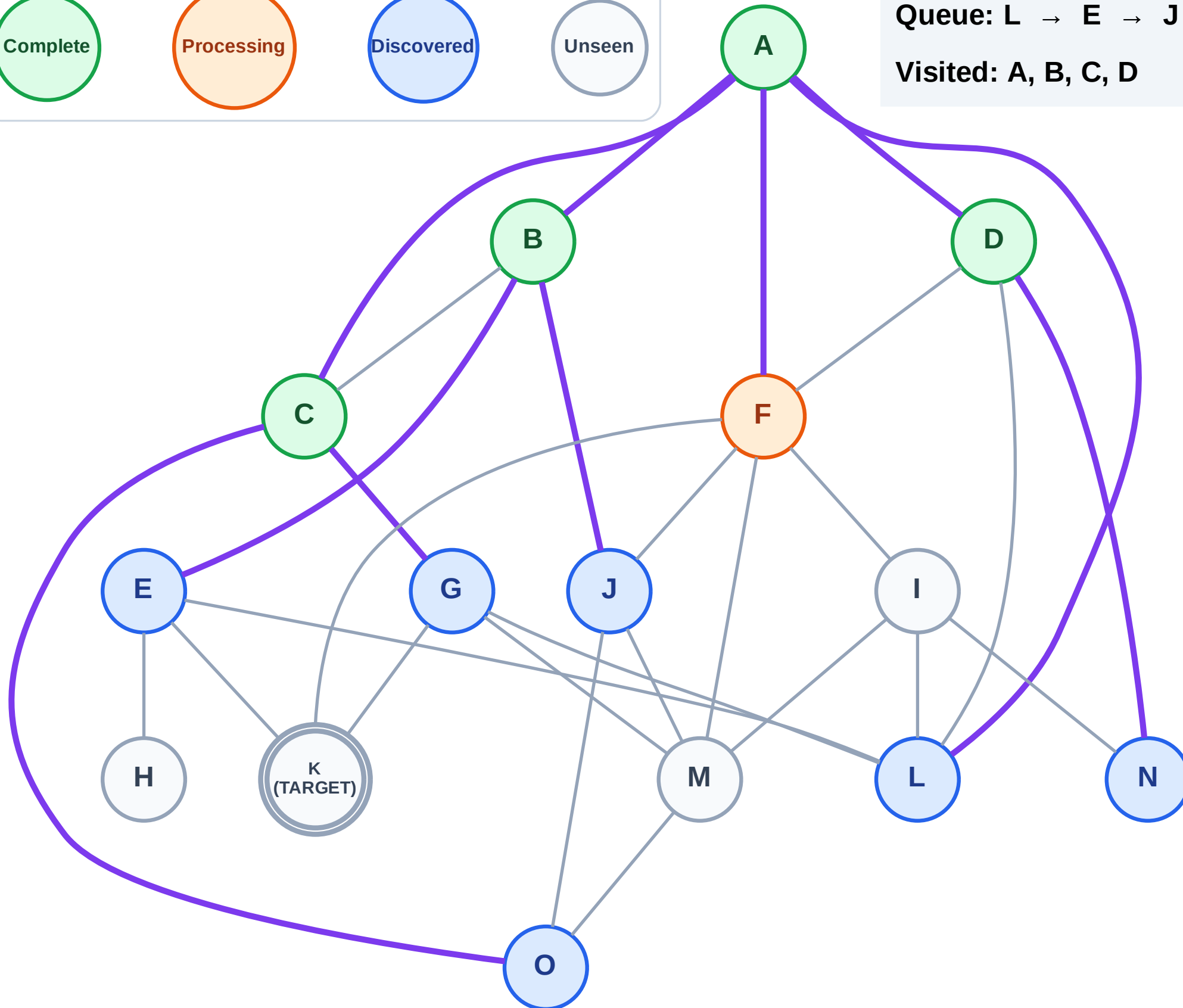


Step 10: Process node F

Legend



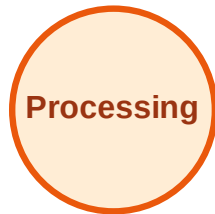
Visited: A, B, C, D



BFS Traversal

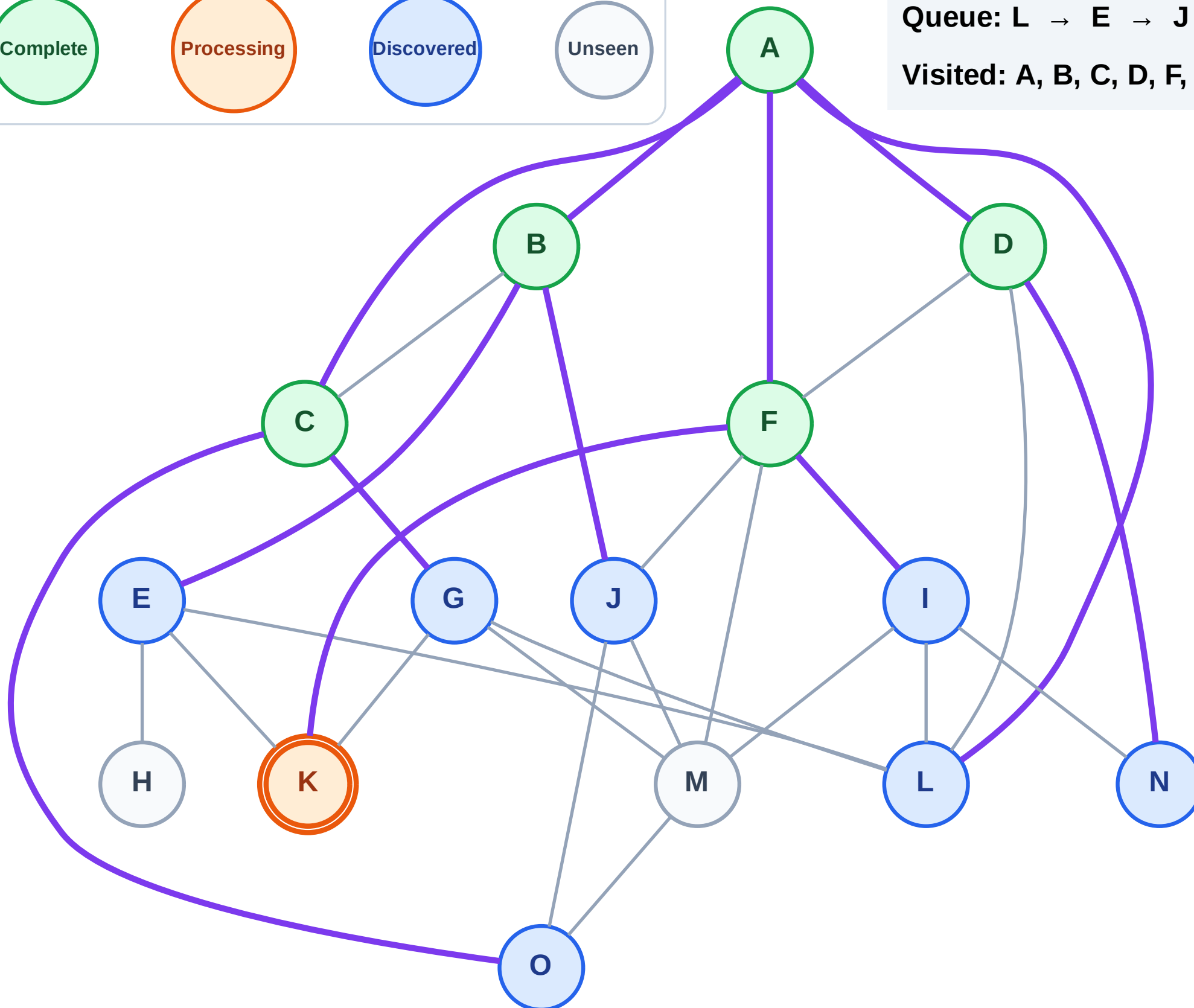
Step 11: TARGET NODE K FOUND (SHORT-CIRCUITED)! Path (2 hops): A → F → K

Legend



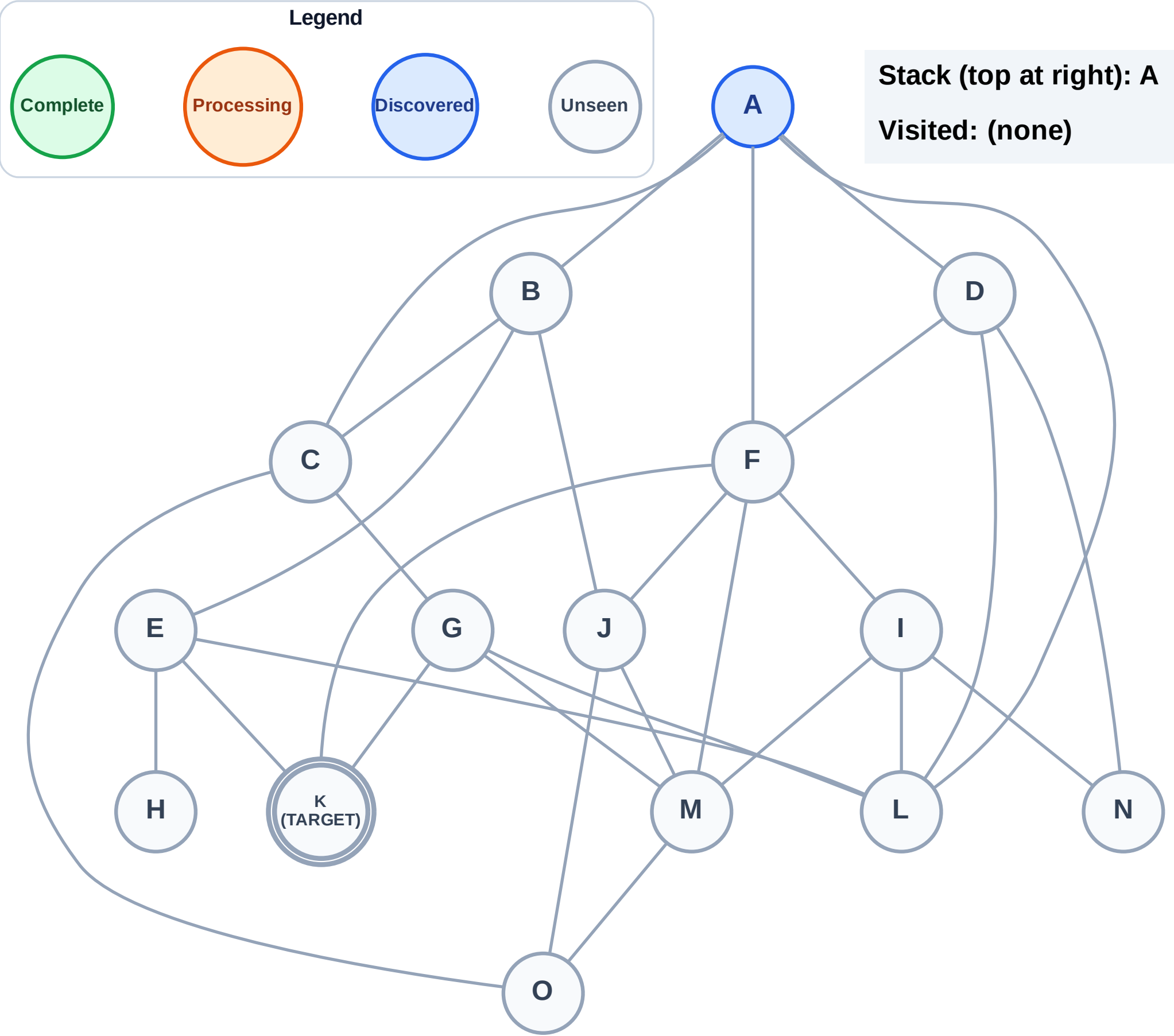
Queue: L → E → J → G → O → N → I

Visited: A, B, C, D, F, K



DFS Traversal

Step 1: Start at A (Seeking target node K (Eager Short-Circuit))



DFS Traversal

Step 2: Process node A

Legend

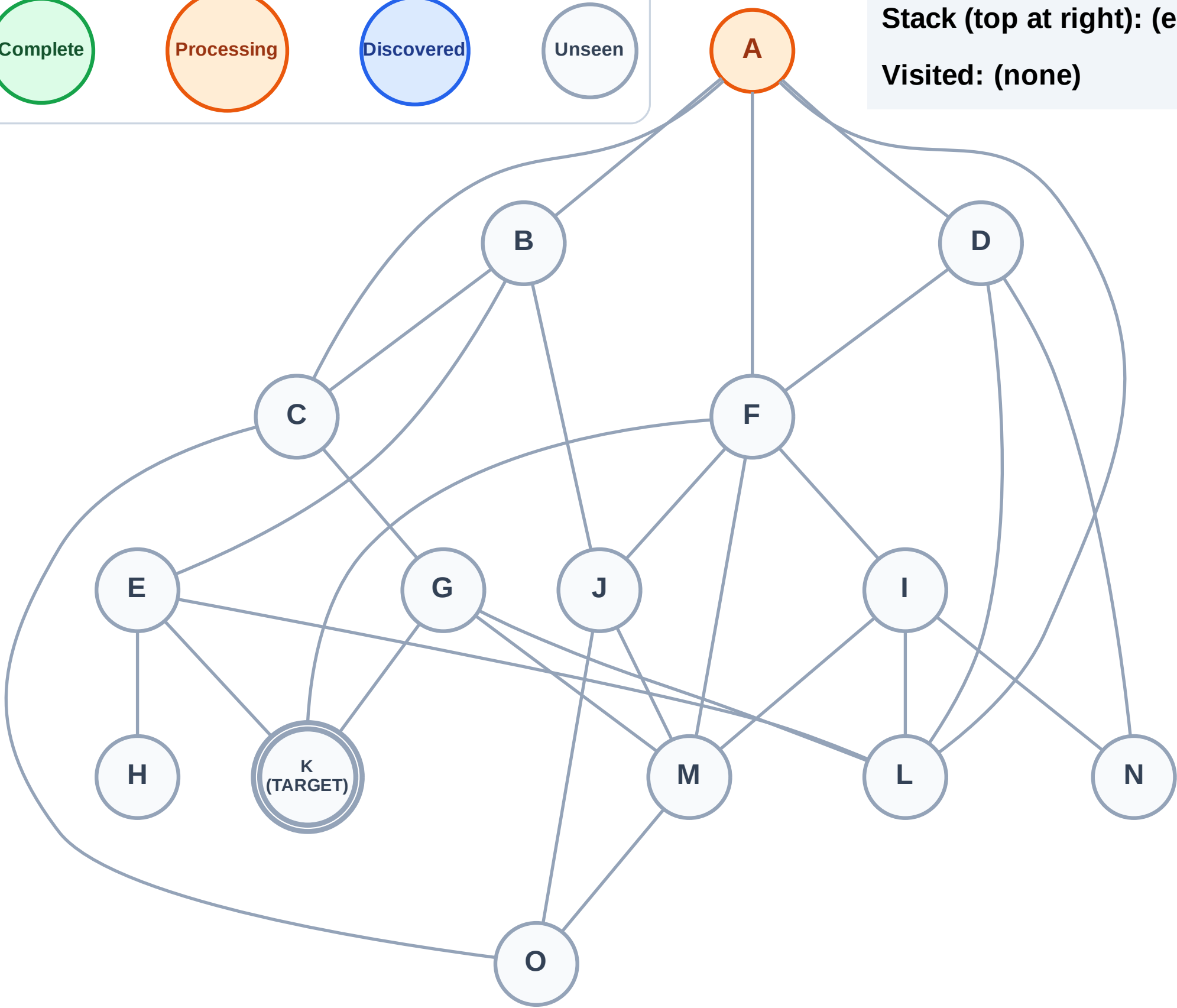
Complete

Processing

Discovered

Unseen

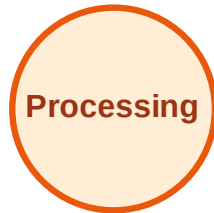
Stack (top at right): (empty)
Visited: (none)



DFS Traversal

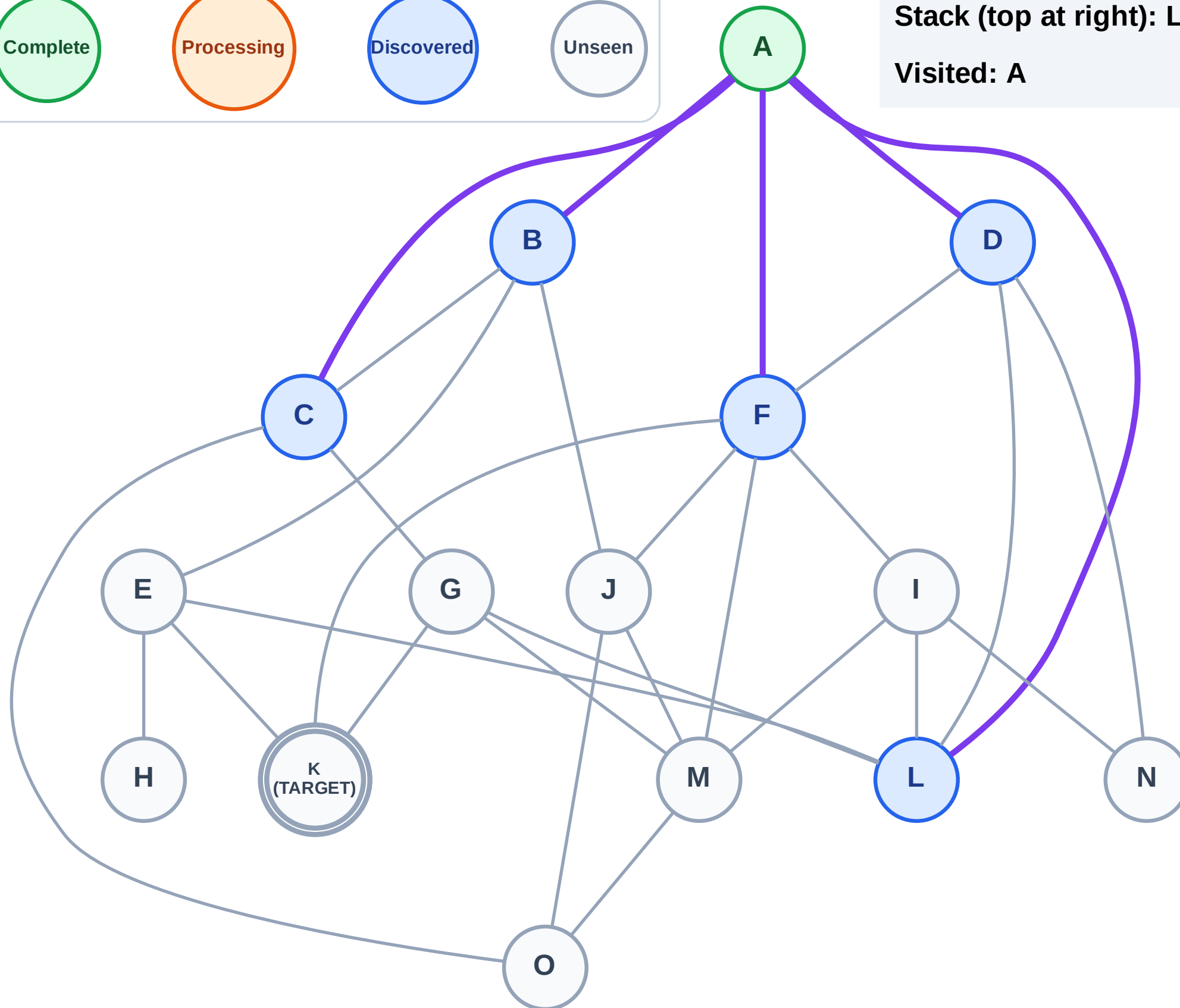
Step 3: Discover L, F, D, C, B from A

Legend



Stack (top at right): L | F | D | C | B

Visited: A



DFS Traversal

Step 4: Process node B

Legend

Complete

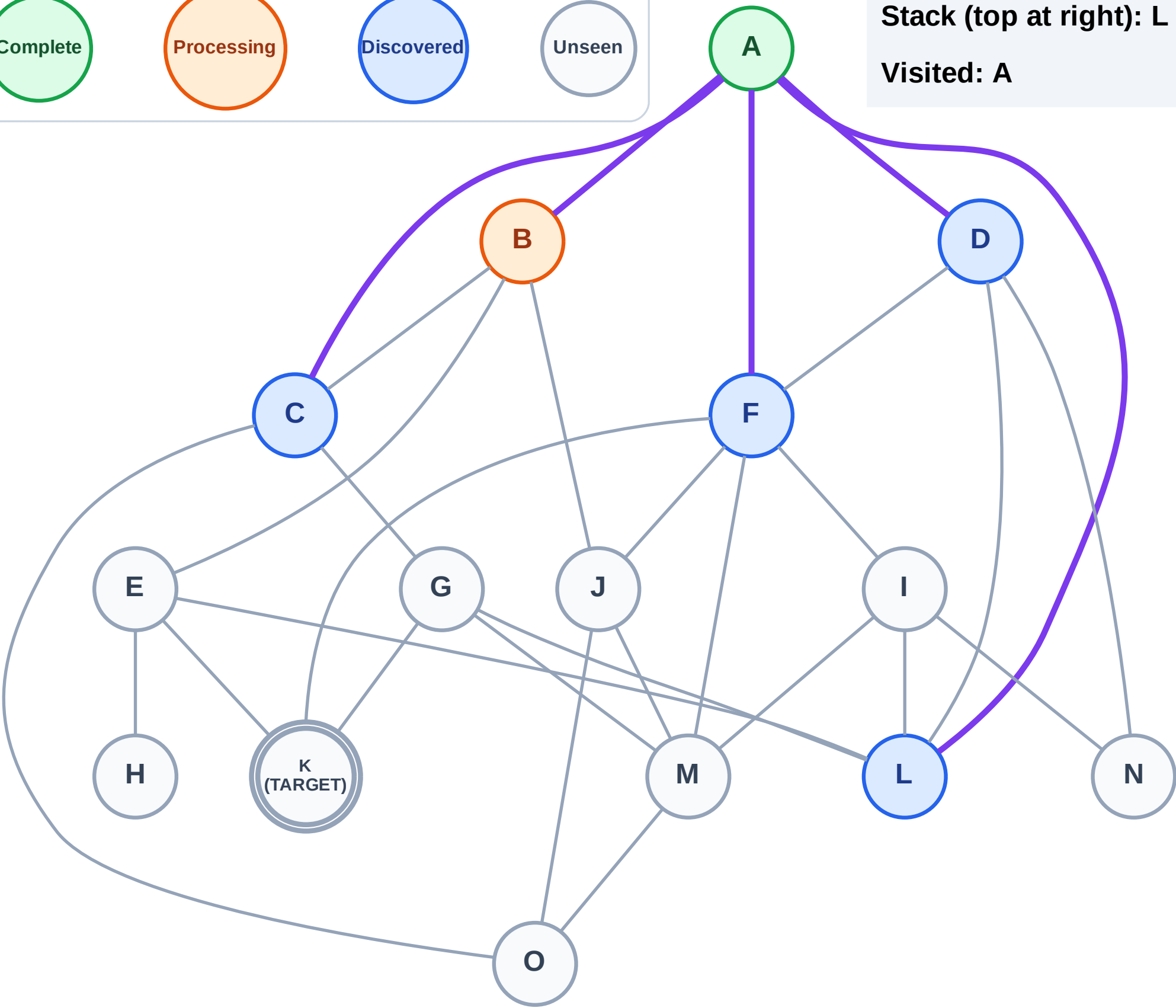
Processing

Discovered

Unseen

Stack (top at right): L | F | D | C

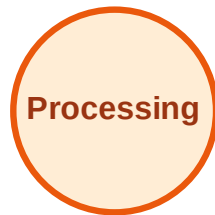
Visited: A



DFS Traversal

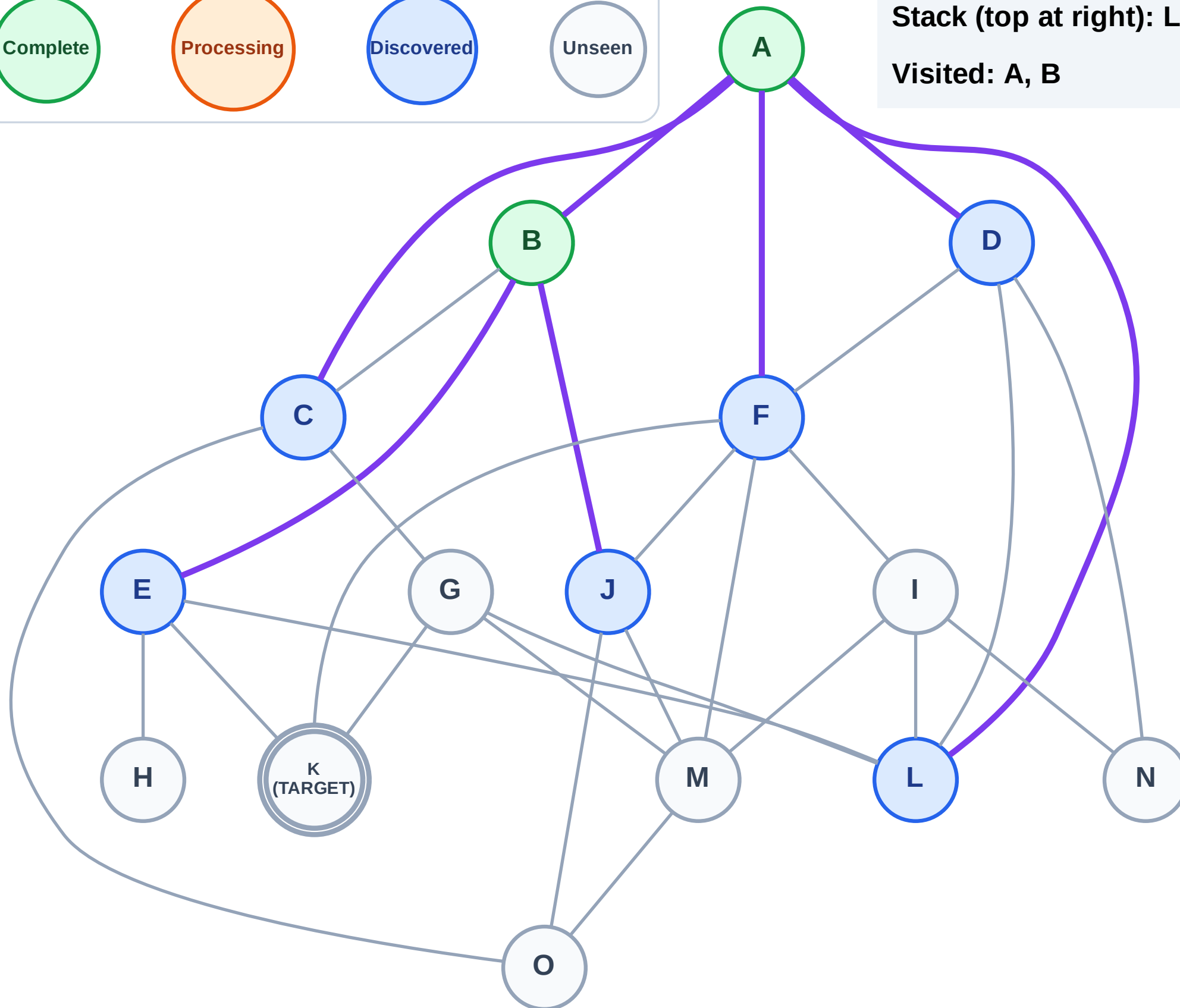
Step 5: Discover J, E from B

Legend



Stack (top at right): L | F | D | C | J | E

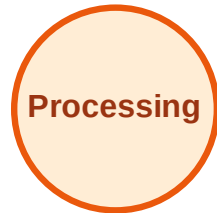
Visited: A, B



DFS Traversal

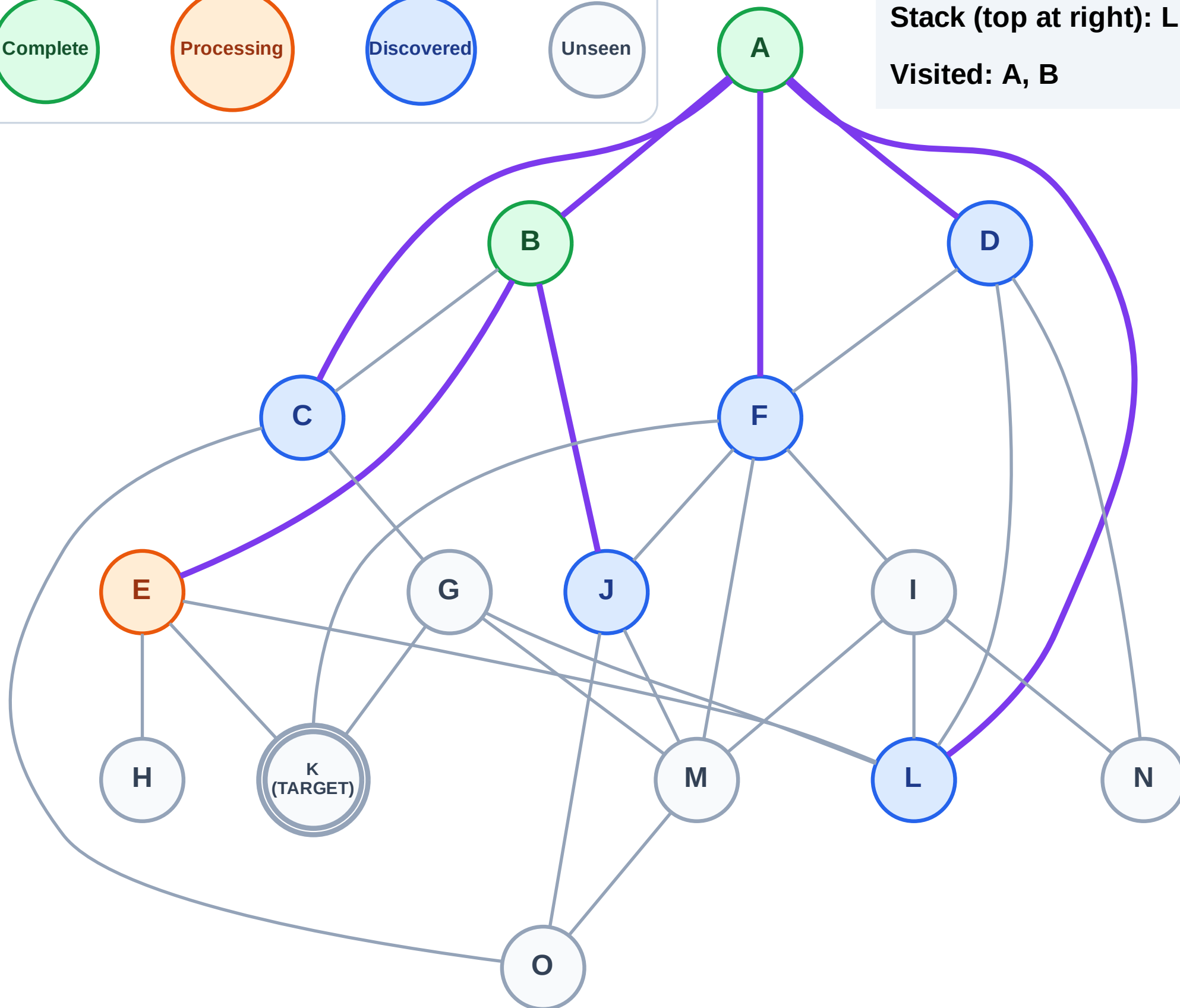
Step 6: Process node E

Legend



Stack (top at right): L | F | D | C | J

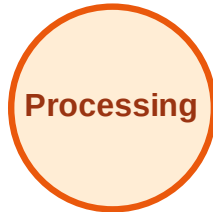
Visited: A, B



DFS Traversal

Step 7: TARGET NODE K FOUND (SHORT-CIRCUITED)! Path (3 hops): A → B → E → K

Legend



Stack (top at right): L | F | D | C | J

Visited: A, B, E, K

